

Pacific Biosciences

NASDAQ : PACB · \$420M mkt cap · 310M sh · \$276M cash, \$644M debt · \$0.276B cash vs ~\$0.644B convertible notes (due 2029/2030) → ~\$0.368B net debt > the equity cap; FY25 revenue \$160M (+4%): consumables \$82M (+16%), instruments \$53.8M (-18%); Cash-flow-positive targeted 2027; FY26 opex guided \$220-225M; ~310M shares, no reverse split

\$1.36

THE CENTRAL QUESTION

At \$1.32, can long-read consumables reach cash-flow breakeven before the ~\$644M of converts mature in 2029/2030 — or does the net debt swamp the equity first?

Pacific Biosciences sells long-read HiFi DNA sequencers (Revio, Vega) and the high-margin consumables that run on them — the #3 behind Illumina's short-read scale and Oxford Nanopore in long-read. The \$1.32 sticker hides the setup: \$0.276B cash against ~\$0.644B of convertible notes is ~\$0.368B net debt, larger than the \$410M equity cap — a financially-levered setup. The DCF asks whether ~16%-growing consumable pull-through reaches cash-flow positive (targeted 2027) before the converts mature. Roughly half the probability mass is worth ~\$0 because the debt outranks the equity; nearly all the value sits in the ~18% long-read-wins tail. Weighted \$1.72 vs spot \$1.32 (+31%) — a cheap levered option, not a value stock.

PROBABILITY-WEIGHTED EXPECTED VALUE

\$1.72

expected fair value today
+26.7% vs spot \$1.36

FORWARD COMPOUNDED VALUE

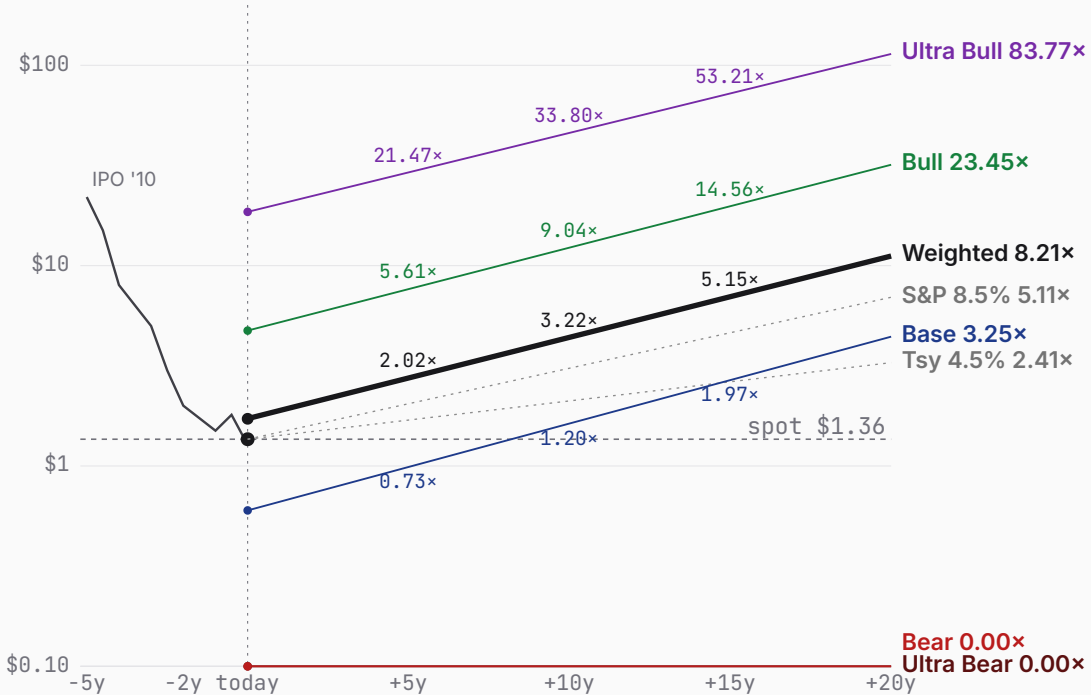
+5y	+10y	+15y	+20y
\$2.75	\$4.39	\$7.00	\$11.17
2.02× spot	3.22× spot	5.15× spot	8.21× spot

WEIGHTING RATIONALE

- Ultra Bear 18% Long-read stalls; converts wipe out equity.
- Bear 34% Growth too slow to clear the debt in time.
- Base 30% Breakeven ~2027; equity just clears converts.
- Bull 13% Long-read inflects; consumables beat the debt.
- Ultra Bull 5% Long-read wins big WGS share; a re-rating.

Bull (13%) + Ultra Bull (5%) together contribute \$1.54 — 90% of the \$1.72 expected value despite 18% combined probability. Today's spot (\$1.36) sits 21% below the weighted expected; forward-weighted value crosses spot between +5y and +10y.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED FORWARD



ULTRA BEAR	\$0.00	BEAR	\$0.00	BASE	\$0.60	BULL	\$4.74	ULTRA BULL	\$18.55
	-100.0% vs spot		-100.0% vs spot		-55.9% vs spot		+248.5% vs spot		+1264.0% vs spot
TAM 3.0% P(fail) 50% S2C 1.4 Prob 18%		TAM 6.1% P(fail) 35% S2C 1.5 Prob 34%		TAM 7.8% P(fail) 22% S2C 1.6 Prob 30%		TAM 13.1% P(fail) 12% S2C 1.8 Prob 13%		TAM 23.8% P(fail) 6% S2C 2.0 Prob 5%	
Long-read stalls; equity wiped		Growth too slow for the debt		Breakeven ~2027; equity clears		Long-read inflects; debt covered		Long-read wins WGS; re-rating	



Pacific Biosciences scenario narratives

PROBABILITY WEIGHTS

Ultra Bear 18% Bear 34% Base 30% Bull 13% Ultra Bull 5% · Weighted expected \$1.72 (+26.7% vs spot)

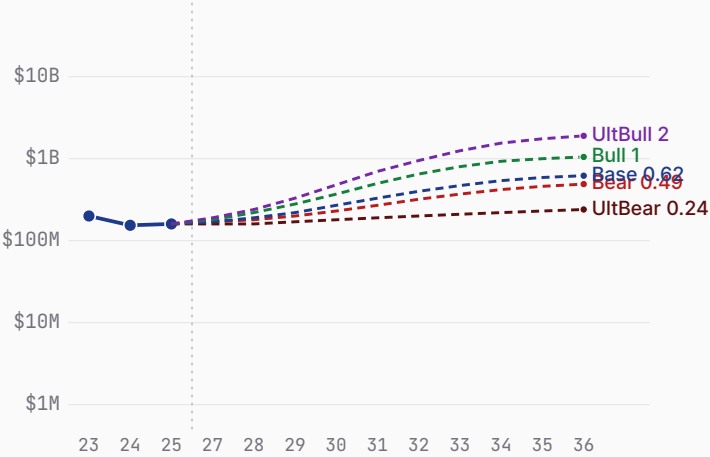
ULTRA BEAR	BEAR	BASE	BULL	ULTRA BULL
\$0.00	\$0.00	\$0.60	\$4.74	\$18.55
-100.0% vs spot	-100.0% vs spot	-55.9% vs spot	+248.5% vs spot	+1264.0% vs spot
Probability 18%	Probability 34%	Probability 30%	Probability 13%	Probability 5%
Long-read stalls; equity wiped	Growth too slow for the debt	Breakeven ~2027; equity clears	Long-read inflects; debt covered	Long-read wins WGS; re-rating
WHY 18% The razor/blade pivot is real but unproven at scale, and a 50% p_fail reflects a sub-scale #3 carrying net debt above its equity cap into a 2029/2030 maturity wall — plausible but not the modal path. WHAT HAPPENS Long-read adoption stalls and instrument placements keep shrinking, leaving FY35 revenue near \$0.24B at a thin 6% operating margin. Pull-through never funds the cost base, and burn forces small dilutive raises that add shares at a low price while the value migrates to the noteholders. With a 50% failure probability the ~\$0.644B of converts outrank a tiny equity stub, so common is worth ~\$0 in restructuring (distress \$0.30/share). DCF -\$1.06; probability-weighted contribution rounds to \$0. The debt, not the business, sets the floor.	WHY 34% The largest single bucket. Reflects the central tension — real but unspectacular long-read uptake against a fixed debt wall — where survival is likely but equity value is not, given how little cushion sits above the converts. WHAT HAPPENS Consumables grow modestly to ~\$0.49B FY35 revenue at a 19% margin, but the ramp is too slow to cover the converts before they mature. The business limps toward breakeven while the installed base compounds only gradually and instruments stay soft. At 35% p_fail the equity barely exists once ~\$0.644B of debt is serviced or refinanced into dilution; DCF -\$0.35, weighted contribution ~\$0. A workable product that simply can't out-run its own balance sheet in time.	WHY 30% Requires the consumables flywheel to fund operations on roughly the guided timeline and the converts to be covered or refinanced cleanly. A demanding but reachable path given the restructuring already underway and the 2027 breakeven target. WHAT HAPPENS Steady long-read adoption plus the FY26 cost cuts (opex \$220-225M, headcount ~16%) reach cash-flow breakeven around 2027. FY35 revenue ~\$0.62B at a 21% margin as Revio pull-through (~\$229K annualized) and a growing Vega base compound high-margin consumables. At 22% p_fail the equity barely clears the ~\$0.644B of converts rather than being consumed by them; DCF \$0.62, weighted contribution \$0.55. The modal case is survival with thin positive equity — close to today's price, which is why the headline lands near fair.	WHY 13% Needs durable consumable pull-through growth and a clinical/research shift toward long-read accuracy that PACB captures as the #3 — a real but lower-probability inflection, hence 13%. WHAT HAPPENS Long-read genuinely inflects: the Revio and Vega base compounds high-margin consumables to ~\$1.05B FY35 revenue at a 26% margin. Operating leverage turns the razor/blade model cash-generative well ahead of the maturity wall. At 12% p_fail the converts are comfortably covered and the equity captures the upside above them; DCF \$5.34, weighted contribution \$4.74. This is the first scenario where financial leverage works for the common holder rather than against it.	WHY 5% The lowest-probability outcome: long-read displacing short-read at population scale against Illumina's entrenched economics. Low odds, but the operating and financial leverage make the payoff large enough to matter to the weighted value. WHAT HAPPENS Long-read takes a large share of whole-genome sequencing and PACB rides ~12x revenue growth to ~\$1.9B FY35 at a 31% margin. The installed base becomes a consumables annuity and the equity re-rates as the debt shrinks to a rounding error against cash flow. At 6% p_fail the converts are trivially covered; DCF \$19.71, weighted contribution \$18.55. The fat tail that, alongside the bull, carries the distribution and offsets the ~50% of mass worth ~\$0.

Pacific Biosciences business snapshot

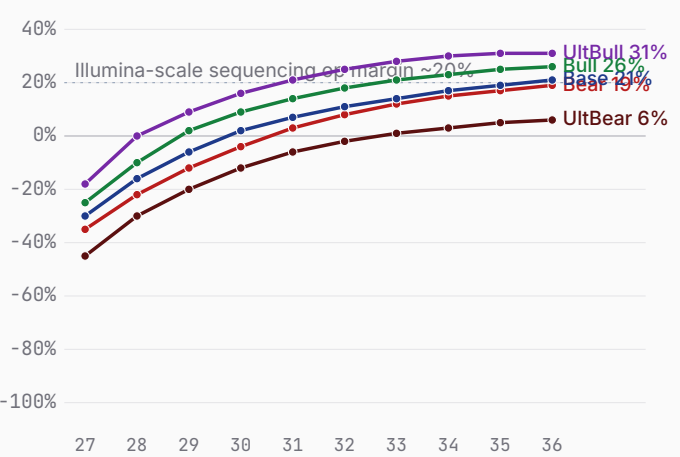
Bear \$0.00 Base \$0.60 Bull \$4.74

FY23-FY25 history + FY26-FY35 scenario projections · fiscal years end Dec 31 · Q1'26 (May 2026) release

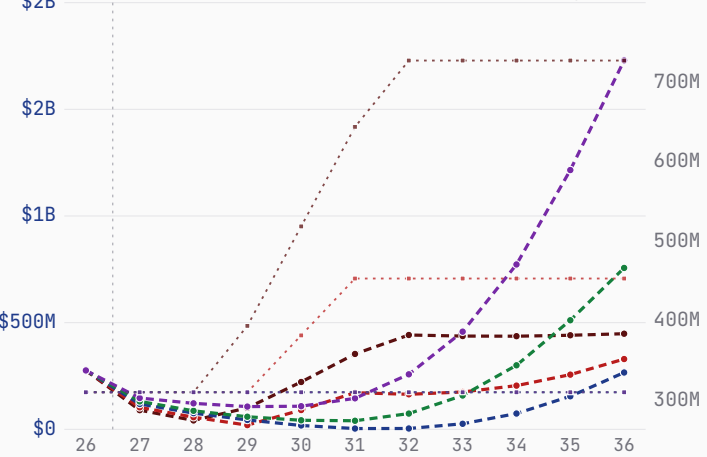
Revenue — history + scenarios to FY36 (log)



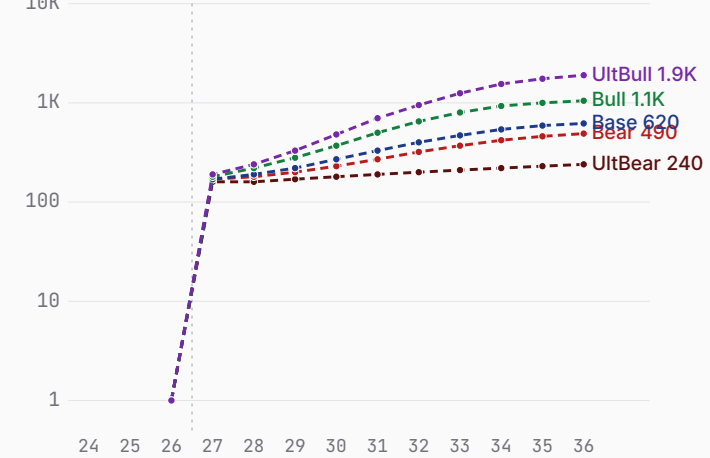
Path to profitability — op margin (FY27→FY36)



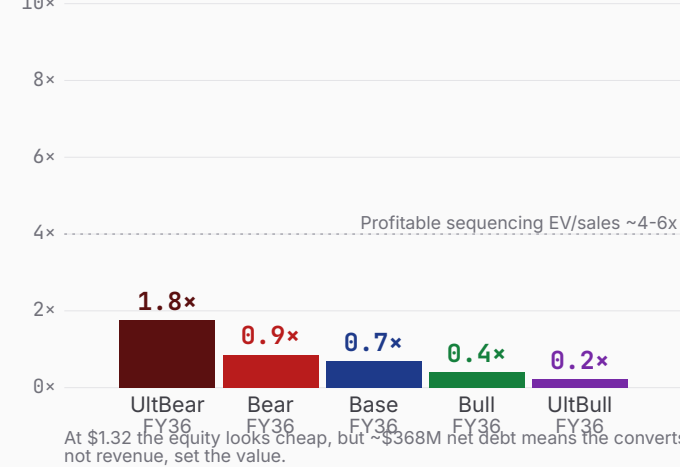
Cash + dilution (FY26 → FY36)



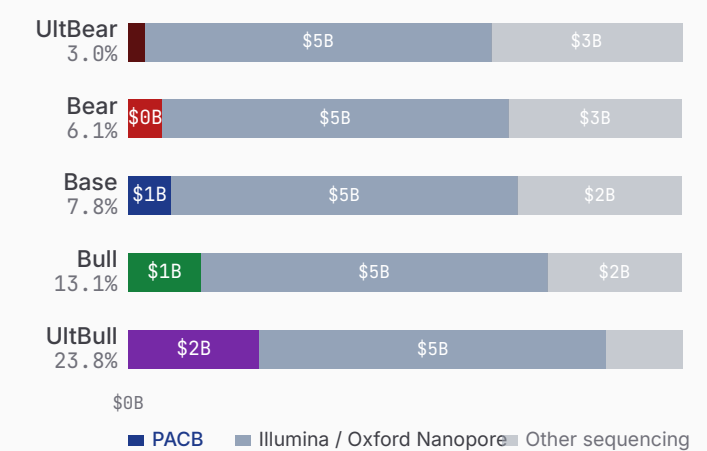
Revenue ramp (log)



\$0.42B mkt cap as P/S on FY36 rev



\$8B long-read sequencing — PACB share



Sources: PACB FY2025 + Q1'26 results (CIK 1299130), convertible-note terms (1.375% 2030 ~\$441M + 1.50% 2029 ~\$200M), FY26 opex/breakeven guidance. Equity bridge carries the ~\$644M converts as net debt. FCF is NOL-shielded pre-tax over the explicit window. TAM: the long-read serviceable slice of the sequencing market, ~\$8B.

Pacific Biosciences 7 Powers · the moat, scored

Bear \$0.00 Base \$0.60 Bull \$4.74

Arena. Long-read DNA sequencing within the broader sequencing market — PACB (HiFi/SMRT) versus Illumina's dominant short-read scale and Oxford Nanopore in direct long-read.

POWER ORIENTATION
origination window: closing

POWER ORIENTATION · 7 POWERS

Scale Economies	Sub-scale #3 at ~\$160M revenue; Illumina's ~\$4.5B base buys cost and R&D advantages PACB can't match.
Network Economies	Some ecosystem pull from published HiFi methods and installed-base data, but no strong cross-side network.
Counter-Positioning	Long-read is structurally different from short-read, but Illumina can and does add long-read offerings, blunting the wedge.
Switching Costs	Installed Revio/Vega plus validated lab workflows create real stickiness, anchoring recurring consumable pull-through.
Branding	Credible accuracy reputation in long-read, but no pricing-power brand against the incumbent.
Cornered Resource · thesis leans here	HiFi/SMRT IP and accuracy are the asset PACB is trying to turn into Power, but contested by Nanopore and Illumina.
Process Power	Manufacturing and chemistry know-how exist; not yet a durable, hard-to-replicate process advantage at scale.

COMPETITIVE READ

A Power is being attempted, not yet originated. PACB is trying to convert HiFi/SMRT accuracy into a cornered-resource and process Power in long-read, but it lags Illumina's scale economies badly and shares the long-read field with Oxford Nanopore. The falsifier the bull must clear is observable and singular — consumable pull-through compounding fast enough to reach cash-flow breakeven and clear the ~\$644M convert wall before 2029/2030. Miss it and the leverage, not the technology, decides the outcome.

THE RACE · NAMED RIVALS

Illumina	strong cash flow, self-funded
~70% terminal share · Short-read scale leader, ~\$4.5B revenue; dominant economics, adding long-read	
Oxford Nanopore	public, funded but pre-profit
~12% terminal share · Direct long-read rival (LSE); portable, real-time reads	
Pacific Biosciences	net debt ~\$368M; burning cash
~18% terminal share · The #3; HiFi accuracy edge, net-debt-constrained	

LEAD / LAG · FALSIFIERS

Long-read accuracy (HiFi)	Leading on accuracy vs Oxford Nanopore	LEADING
Scale / cost per genome	Sub-scale vs Illumina	LAGGING
Balance-sheet strength	Net debt above equity cap vs Illumina	LAGGING

Pacific Biosciences show your work

Bear \$0.00 Base \$0.60 Bull \$4.74 · Weighted \$1.72 (+26.7%)

ULTRA BEAR

\$0.00

BEAR

\$0.00

BASE

\$0.60

BULL

\$4.74

ULTRA BULL

\$18.55

ASSUMPTIONS

TAM Year-10 (\$B)	8
Mature share (%)	3.0
Mature op margin (%)	6
Sales-to-capital	1.4
Terminal WACC (%)	11.5
Terminal growth (%)	3.0
P(failure) (%)	50
Scenario probability (%)	18

10-YEAR DCF · \$B

FY	Rev	Margin	FCF	PV FCF
FY27	0.16	-45%	-0.07	-0.06
FY28	0.16	-30%	-0.05	-0.04
FY29	0.17	-20%	-0.04	-0.03
FY30	0.18	-12%	-0.03	-0.02
FY31	0.19	-6%	-0.02	-0.01
FY32	0.20	-2%	-0.01	-0.01
FY33	0.21	+1%	-0.01	-0.00
FY34	0.22	+3%	-0.00	+0.00
FY35	0.23	+5%	+0.00	+0.00
FY36	0.24	+6%	+0.01	+0.00

Σ PV explicit FCF -0.16

+ PV terminal (g 3.0%) 0.03

EQUITY BUILD · \$B & PER SHARE

Operating EV	\$-0.13B
+ Cash – total debt	-\$0.37B
= Equity value	\$-0.50B
Raised over period	\$0.50B
Dilution by FY36	+134%
FY36 shares (M)	727

DCF per share	\$-0.69
× (1-P_fail) [50%]	\$-0.34
+ P_fail × distress [50% × \$0.30]	\$0.15

= Expected per share \$0.00

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
\$0.00	\$0.00	\$0.00	\$0.00
0.00×	0.00×	0.00×	0.00×

ASSUMPTIONS

TAM Year-10 (\$B)	8
Mature share (%)	6.1
Mature op margin (%)	19
Sales-to-capital	1.5
Terminal WACC (%)	11.0
Terminal growth (%)	3.5
P(failure) (%)	35
Scenario probability (%)	34

10-YEAR DCF · \$B

FY	Rev	Margin	FCF	PV FCF
FY27	0.17	-35%	-0.07	-0.06
FY28	0.18	-22%	-0.05	-0.04
FY29	0.20	-12%	-0.04	-0.03
FY30	0.23	-4%	-0.03	-0.02
FY31	0.27	+3%	-0.02	-0.01
FY32	0.32	+8%	-0.01	-0.00
FY33	0.37	+12%	+0.01	+0.01
FY34	0.42	+15%	+0.03	+0.01
FY35	0.46	+17%	+0.05	+0.02
FY36	0.49	+19%	+0.07	+0.02

Σ PV explicit FCF -0.09

+ PV terminal (g 3.5%) 0.32

EQUITY BUILD · \$B & PER SHARE

Operating EV	\$0.23B
+ Cash – total debt	-\$0.37B
= Equity value	\$-0.14B
Raised over period	\$0.20B
Dilution by FY36	+46%
FY36 shares (M)	453

DCF per share	\$-0.31
× (1-P_fail) [65%]	\$-0.20
+ P_fail × distress [35% × \$0.30]	\$0.10

= Expected per share \$0.00

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
\$0.00	\$0.00	\$0.00	\$0.00
0.00×	0.00×	0.00×	0.00×

ASSUMPTIONS

TAM Year-10 (\$B)	8
Mature share (%)	7.8
Mature op margin (%)	21
Sales-to-capital	1.6
Terminal WACC (%)	10.5
Terminal growth (%)	4.0
P(failure) (%)	22
Scenario probability (%)	30

10-YEAR DCF · \$B

FY	Rev	Margin	FCF	PV FCF
FY27	0.17	-30%	-0.06	-0.05
FY28	0.19	-16%	-0.04	-0.03
FY29	0.22	-6%	-0.03	-0.02
FY30	0.27	+2%	-0.03	-0.02
FY31	0.33	+7%	-0.01	-0.01
FY32	0.40	+11%	+0.00	+0.00
FY33	0.47	+14%	+0.02	+0.01
FY34	0.54	+17%	+0.05	+0.02
FY35	0.59	+19%	+0.08	+0.03
FY36	0.62	+21%	+0.11	+0.04

Σ PV explicit FCF -0.03

+ PV terminal (g 4.0%) 0.61

EQUITY BUILD · \$B & PER SHARE

Operating EV	\$0.58B
+ Cash – total debt	-\$0.37B
= Equity value	\$0.21B
Raised over period	\$0.00B
Dilution by FY36	+0%
FY36 shares (M)	311

DCF per share	\$0.68
× (1-P_fail) [78%]	\$0.53
+ P_fail × distress [22% × \$0.30]	\$0.07

= Expected per share \$0.60

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
\$0.99	\$1.63	\$2.68	\$4.42
0.73×	1.20×	1.97×	3.25×

ASSUMPTIONS

TAM Year-10 (\$B)	8
Mature share (%)	13.1
Mature op margin (%)	26
Sales-to-capital	1.8
Terminal WACC (%)	10.0
Terminal growth (%)	5.0
P(failure) (%)	12
Scenario probability (%)	13

10-YEAR DCF · \$B

FY	Rev	Margin	FCF	PV FCF
FY27	0.18	-25%	-0.06	-0.05
FY28	0.22	-10%	-0.04	-0.04
FY29	0.28	+2%	-0.03	-0.02
FY30	0.37	+9%	-0.02	-0.01
FY31	0.50	+14%	-0.00	-0.00
FY32	0.65	+18%	+0.03	+0.02
FY33	0.80	+21%	+0.09	+0.04
FY34	0.93	+23%	+0.14	+0.06
FY35	1.00	+25%	+0.21	+0.08
FY36	1.05	+26%	+0.24	+0.09

Σ PV explicit FCF +0.18

+ PV terminal (g 5.0%) 1.85

EQUITY BUILD · \$B & PER SHARE

Operating EV	\$2.03B
+ Cash – total debt	-\$0.37B
= Equity value	\$1.66B
Raised over period	\$0.00B
Dilution by FY36	+0%
FY36 shares (M)	311

DCF per share	\$5.34
× (1-P_fail) [88%]	\$4.70
+ P_fail × distress [12% × \$0.30]	\$0.04

= Expected per share \$4.74

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
\$7.63	\$12.29	\$19.80	\$31.89
5.61×	9.04×	14.56×	23.45×

ASSUMPTIONS

TAM Year-10 (\$B)	8
Mature share (%)	23.8
Mature op margin (%)	31
Sales-to-capital	2.0
Terminal WACC (%)	9.5
Terminal growth (%)	6.0
P(failure) (%)	6
Scenario probability (%)	5

10-YEAR DCF · \$B

FY	Rev	Margin	FCF	PV FCF
FY27	0.19	-18%	-0.05	-0.04
FY28	0.24	+0%	-0.03	-0.02
FY29	0.33	+9%	-0.01	-0.01
FY30	0.48	+16%	+0.00	+0.00
FY31	0.70	+21%	+0.04	+0.02
FY32	0.95	+25%	+0.11	+0.06
FY33	1.25	+28%	+0.20	+0.10
FY34	1.55	+30%	+0.32	+0.14
FY35	1.75	+31%	+0.44	+0.18
FY36	1.90	+31%	+0.51	+0.19

Σ PV explicit FCF +0.63

+ PV terminal (g 6.0%) 5.87

EQUITY BUILD · \$B & PER SHARE

Operating EV	\$6.50B
+ Cash – total debt	-\$0.37B
= Equity value	\$6.13B
Raised over period	\$0.00B
Dilution by FY36	+0%
FY36 shares (M)	311

DCF per share	\$19.71
× (1-P_fail) [94%]	\$18.53
+ P_fail × distress [6% × \$0.30]	\$0.02

= Expected per share \$18.55

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
\$29.20	\$45.97	\$72.37	\$113.93
21.47×	33.80×	53.21×	83.77×

Pacific Biosciences

People · Opportunity · Context · Deal

Bear \$0.00 Base \$0.60 Bull \$4.74

Underwrite the position the way a venture investor underwrites a round — across all four legs, public or private. **People** is the leg scored here (observable only); Opportunity, Context and Deal cross-reference the rest of the memo.

PEOPLE · WHO RUNS / OWNS / FUNDS IT

Christian O. Henry

6 yrs in seat

1%insider ownership

LOWkey-person risk

Capital allocation (realized)
CEO since September 2020 (prior: Illumina executive 2005-2017 — CFO, Chief Commercial Officer, GM of Life Sciences). The capital priority is reaching cash-flow breakeven, targeted 2027: FY25 revenue \$160M (+4%), a ~16% headcount reduction and FY26 opex guided to \$220-225M, run against ~\$276M cash versus ~\$644M of convertible notes due 2029/2030 (~\$368M net debt). No dividend or buyback — cash is being preserved to fund the consumables ramp and clear the convert wall.

Incentive alignment
Insiders hold under 1% of the company; the CEO is the largest individual holder at roughly 1.2% (~3.6M shares). Alignment runs through equity awards rather than a founding stake, and the thin insider holding is itself a flag — limited owner-operator skin in the game against the balance-sheet risk.

GOVERNANCE FLAGS

- single class of common stock, one vote per share — no dual-class or super-voting
- combined Chairman and CEO (Henry holds both), with a lead independent director as the counterweight
- board expanded to 10 members in March 2026; largest holder is BlackRock (~7.7%), no founder control block
- levered balance sheet: ~\$644M convertible notes due 2029/2030 exceed the modeled equity value — the central capital-allocation overhang

PEOPLE READ

Professional, experienced management (CEO Henry is ex-Illumina) on a clean single-class one-vote register with a majority-independent 10-member board — the governance positives. The offsets are a combined Chair/CEO, insiders below 1% (thin owner-operator alignment), and a levered balance sheet where the converts exceed the modeled equity, so capital allocation is dominated by the race to breakeven before 2029/2030. The technology is institutionalized, so key-person risk is low.

THE FOUR LEGS

People

observable composite · 1–5

Opportunity

The scenario distribution · the business snapshot

Context

The 7 Powers analysis (Power Origination)

Deal

Open-market common — entry price weighed against the probability-weighted fair value (the +30.6% finding) + position sizing.



Pacific Biosciences supporting analysis · disclaimers · glossary

Bear \$0.00 Base \$0.60 Bull \$4.74

PUSHBACK · WHY THE BASE CASE IS TOO HARSH

- 1 **Converts were already refinanced**

The 2028 SoftBank wall was pushed to 2029/2030, buying runway — but ~\$644M still outranks a \$410M equity cap and must be cleared.
- 2 **Consumables are inflecting now**

Consumables grew +16% to \$82M while instruments fell -18% — the razor/blade pivot is visibly underway, the bull's required first step.
- 3 **Restructuring de-risks the burn**

FY26 opex guided down to \$220-225M with headcount -16% targets cash-flow-positive in 2027 — if hit, base-case equity survives.
- 4 **Long-read accuracy is a real edge**

HiFi reads solve problems short-read can't; if whole-genome sequencing shifts to long-read, the #3 still captures a growing consumables annuity.
- 5 **It is a barbell, not sticker-cheap**

The \$1.32 looks cheap, but net debt above the equity cap means ~half the mass is ~\$0 — the price reflects leverage, not a bargain.

FALSIFICATION TRIGGERS

Bull validation Consumable pull-through accelerates above ~\$229K/Revio; adj-EBITDA turns positive; the FY27 cash-flow-positive target is reaffirmed; converts refinanced non-dilutively.	Bear validation Instrument declines persist; pull-through flattens; burn stays ~\$45M/qtr; a dilutive equity or convert refinancing before 2029.	Reframe needed A larger-than-expected raise, a covenant or maturity event, or a credible buyer for the long-read franchise — any of which resets the equity thesis.
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GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

long-read / HiFi sequencing — DNA sequencing that reads long, highly accurate fragments (HiFi), resolving regions short-read instruments struggle with.	consumable pull-through — Recurring high-margin reagent/kit revenue each installed instrument generates per year — the 'blade' in the razor/blade model.	Revio — PACB's high-throughput long-read sequencer (346 installed); the main pull-through driver at ~\$229K annualized.
convertible notes — Debt that can convert to equity; ~\$644M here (due 2029/2030) that outranks common holders until repaid or converted.	razor/blade model — Selling instruments (razor) to drive recurring consumable sales (blade); the pivot underway as consumables grow and instruments shrink.	p_fail — Modeled probability the equity is wiped out (distress ~\$0.30/share) before the scenario's terminal year — elevated here by net debt above the equity cap.